

Broker, Real-Time Intelligence & Execution Blueprint

A practical architecture for an event-aware, continuously updating currency trading assistant

Recommended first broker: OANDA (U.S.)

Core principle: The system should behave like a disciplined professional analyst, not like a gambler. It continuously observes prices, scheduled macro releases, central-bank information and relevant news; updates its view as evidence changes; and only permits a trade when the expected edge survives independent risk checks. It should be allowed to say NO TRADE.

IMPORTANT

This is an engineering and research blueprint, not a promise of profit. Leveraged forex can produce rapid losses. The initial deployment should be paper/demo only until the system has passed backtesting, walk-forward testing and an extended live-paper evaluation.

1. Executive Summary

This document specifies a first-generation AI-assisted forex platform that combines real-time broker prices with macroeconomic releases, central-bank information and current-affairs/news signals. The recommended starting broker is OANDA because its U.S. platform explicitly supports API trading, offers a demo account, supports small trade sizes, and exposes real-time pricing and order endpoints. [1][2][3]

The system is intentionally split into two layers: an Intelligence Layer that forms and continuously revises a market view, and an Execution/Risk Layer that decides whether that view is safe enough to trade. The intelligence model may recommend BUY, SELL or NO TRADE; the risk engine always has final veto power.

Decision	Recommended choice	Reason
Primary broker	OANDA U.S.	Forex-focused API, real-time pricing stream, programmatic orders, demo path, small unit sizing.
Primary language	Python 3.12+	Strong ecosystem for streaming data, ML/NLP, backtesting and broker integration.
Core pairs for v1	EUR/USD, GBP/USD, USD/JPY, USD/CAD, AUD/USD	Liquid majors with strong macro linkages and abundant data.
Initial mode	Paper/demo only	Validates latency, slippage, model behavior and risk logic without risking capital.
Model style	Hybrid ensemble	Combines price/volatility, macro surprise, event/news sentiment and regime classification.
Decision outputs	BUY / SELL / NO TRADE + probability + reason	Forces explicit uncertainty and prevents compulsive trading.
Risk per live trade (later)	Default 0.25%-0.50%; hard ceiling 1%	Survival and drawdown control matter more than maximizing a single trade.

2. Broker Selection

2.1 Recommended broker: OANDA

For this project, OANDA is the most practical first broker. OANDA states that its v20 REST API gives developers programmatic access to automated trading, real-time market data and historical pricing. Its U.S. account comparison also lists API trading for standard accounts and variable contract sizes as low as one unit. [1][2]

Requirement	OANDA fit	Design implication
Real-time price access	Yes	Use the pricing stream as the authoritative executable-market feed.
API order placement	Yes	Execution service can create market/limit orders and attach stop-loss/take-profit logic.
Historical candles	Yes	Use broker-aligned history for backtests and feature generation.
Demo/paper route	Yes	Mandatory first deployment mode.
Small position sizes	Yes, down to unit-level sizing per account comparison	Useful for tightly controlled live transition.

Forex specialization	Strong	Keeps v1 focused instead of adding stocks/options complexity.
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2.2 API behavior that matters

OANDA documents a streaming pricing endpoint. The stream can provide up to four prices per second per requested instrument, rather than every raw market tick, and sends heartbeats every five seconds. This is fast enough for an event-aware intraday/swing system, but it is not an ultra-high-frequency feed. [3]

OANDA also documents an order endpoint that can create orders programmatically and supports attaching protective stop-loss and take-profit instructions. [4]

Architectural consequence: Do not design this as a microsecond/HFT system. Design it as a fast event-aware decision system operating on seconds-to-hours horizons. The model should refresh rapidly around major events, but it should not assume it receives every interbank tick.

3. Target System Architecture

The model should “behave like a human analyst” in the sense that it gathers evidence from multiple sources, recognizes context, revises beliefs when new facts arrive, and can abstain. It should not imitate human emotion, revenge trading or intuition without evidence.

Layer	Responsibilities
1. Market Data Gateway	OANDA price stream; candles; spread; account status; open positions; order/trade updates.
2. Macro/Event Gateway	Economic calendar; actual vs forecast vs previous; revisions; central-bank decisions; inflation, employment, GDP and rates.
3. News & Current-Affairs Gateway	Fresh relevant headlines; source quality; currency/entity tagging; event classification; novelty and sentiment.
4. Feature Engine	Returns, trend, volatility, spread, correlations, technical structure, event-surprise features, sentiment decay.
5. Regime Detector	Classifies trending/ranging, risk-on/risk-off, high/normal volatility, event shock and liquidity conditions.
6. Prediction Ensemble	Multiple independent models estimate direction, expected move and confidence over defined horizons.
7. Decision Fusion	Combines models with macro/news context; produces BUY/SELL/NO TRADE plus reasons and invalidation conditions.
8. Risk Governor	Position size, maximum loss, drawdown, correlation limits, event lockouts, stale-data checks and kill switch.
9. Execution Service	Submits/cancels orders, attaches stops/targets, reconciles fills and prevents duplicate orders.
10. Monitoring & Audit	Logs every input, model version, prediction, decision, order, fill, error and post-trade outcome.

4. Real-Time Data and Current Affairs

4.1 Market prices

Use OANDA as the execution-price source. The model must always calculate decisions against the broker price/spread it can actually trade, not only a third-party chart. Keep synchronized 5-second, 1-minute, 5-minute, 15-minute, 1-hour, 4-hour and daily feature views. OANDA supports candle granularities starting at five seconds. [3]

4.2 Scheduled macroeconomic data

For economic events, use a calendar service capable of reporting release timestamps, forecasts, previous values, actual values and revisions. Trading Economics documents a near-real-time economic calendar and streaming updates, with actual values sourced from official sources. [5][6]

For validation and slower historical/macro features, favor official sources. FRED exposes economic series programmatically via REST; the U.S. Bureau of Labor Statistics exposes published historical time-series data via API; and the ECB Data Portal exposes statistical data through a RESTful SDMX service. [7][8][9]

Signal family	Examples	How model should use it
Central-bank policy	Fed/ECB/BoE/BoJ decisions, statements, minutes, speeches	Compare stance with market expectations; estimate relative-rate implications.
Inflation	CPI, PCE, HICP	Compute surprise vs consensus and direction of revisions; map to likely rate path.
Labor	NFP/payrolls, unemployment, wages	Assess growth/inflation pressure and rate expectations.
Growth	GDP, PMIs, retail sales, industrial production	Estimate cyclical strength and relative country momentum.
Risk/geopolitics	Conflict, sanctions, elections, fiscal events, major policy announcements	Classify affected currencies; apply source-quality and uncertainty penalties.
Market stress	Volatility jump, spread widening, cross-asset shock	Reduce size or force NO TRADE when liquidity/uncertainty deteriorates.

4.3 News and sentiment

A practical v1 can use Alpha Vantage News & Sentiment as one structured feed because its documentation explicitly supports forex-tagged news and topics such as monetary policy, fiscal policy, macroeconomics and financial markets. [10] For production, consider upgrading to a licensed professional news feed if latency and coverage become critical.

Never let sentiment alone trade: A headline sentiment score is evidence, not a command. The system must check source quality, timestamp, novelty, currency relevance and whether price has already reacted before using a news event in a trade decision.

5. How the Model Should Behave

5.1 Observe → interpret → challenge → decide → monitor

- 1. Observe: continuously ingest prices, spreads, account state, upcoming calendar events and fresh relevant news.
- 2. Interpret: turn raw inputs into structured claims such as “US inflation surprised +0.3 percentage points above consensus” or “BoE statement is more dovish than prior communication.”
- 3. Challenge: run opposing checks. Ask what evidence supports the opposite direction, whether the move is already priced in, and whether sources disagree.
- 4. Decide: output BUY, SELL or NO TRADE with confidence, horizon, proposed entry zone, invalidation point and risk budget.
- 5. Monitor: after entry, continue evaluating the thesis. New evidence can tighten risk, exit the trade or prohibit adding exposure.

5.2 Required decision object

Field	Example
Instrument	EUR_USD
Timestamp (UTC)	2026-08-07T18:31:04Z
Action	BUY / SELL / NO_TRADE
Confidence	0.00-1.00; calibrated, not cosmetic
Horizon	15m / 1h / 4h / 1d
Market regime	Trend / range / shock / high-volatility
Key drivers	Macro surprise; rate differential; price structure; news
Contrary evidence	Explicit list
Entry condition	Price/structure condition, not blind market entry
Invalidation	What must become false for the thesis to fail
Stop distance	Volatility-aware
Target logic	Expected-value / structure based
Max account risk	Risk engine output
Data freshness	Age of every critical feed
Decision explanation	Short machine-readable and human-readable rationale

6. Prediction Model Design

Do not build one giant model that directly maps headlines and candles to BUY/SELL. Use an ensemble with specialized components. This makes failure analysis easier and reduces the chance that one noisy signal controls the account.

Component	Suggested method	Output
Price/technical model	Gradient boosting / temporal model using returns, trend, volatility, ATR, momentum, structure and spread	P(up), expected move, uncertainty
Macro-surprise model	Rule + ML model on actual-vs-consensus surprises, revisions and relative country data	Directional macro score
News-event model	LLM/NLP classifier constrained to structured labels	Affected currencies, direction, severity, novelty, confidence
Regime model	Hidden Markov / clustering / classifier	Trend/range/shock/volatility regime
Cross-market confirmation	Rates/yields, dollar index proxies, commodities where relevant	Confirmation/contradiction score
Meta-model	Logistic/GBM/calibration layer trained only on out-of-sample component outputs	Final calibrated trade probability

6.1 Prediction horizons

Use separate horizon models rather than one vague “next direction” prediction. A 5-minute direction can oppose a 4-hour direction. Recommended v1 horizons: 15 minutes, 1 hour and 4 hours. Add daily only after the intraday system is stable.

6.2 Preventing data leakage

- Every training feature must be timestamped by when it actually became available, not by the period it describes.
- Economic revisions must not leak revised values into historical training examples. Point-in-time data is strongly preferred.
- News must be joined using publication/availability time; never let an article published after a candle influence that candle.
- Use walk-forward validation and untouched final test periods. Random train/test shuffling is inappropriate for time-series trading.
- Backtests must include spread, financing where applicable, latency, slippage assumptions and rejected/partial executions.

7. Trading Decision Logic

The system should be selective. A high-quality NO TRADE decision is a successful output when evidence is weak or risk is abnormal.

Gate	Rule
Freshness gate	Reject decision if broker stream, calendar or required news state is stale beyond configured threshold.
Spread gate	Do not enter when spread is abnormally wide relative to recent distribution.
Event gate	Around tier-1 releases, either use a dedicated event strategy or stay flat. Never let the ordinary strategy unknowingly trade through the release.
Confidence gate	Trade only when calibrated probability and expected value exceed threshold.

Agreement gate	Require minimum agreement across independent components; strong contradiction = NO TRADE.
Risk gate	Position sizing and portfolio limits must pass before order creation.
Duplicate gate	No repeated order caused by duplicate messages/retries.
Kill-switch gate	Any reconciliation failure, data anomaly or daily-loss limit immediately disables new entries.

8. Risk Management: The Non-Negotiable Layer

Priority: The risk engine must be separate from the prediction model. The prediction model is never permitted to increase its own risk limit because it “feels confident.”

Control	Initial paper/live-transition setting
Risk per trade	0.25%-0.50% of equity; absolute ceiling 1%.
Daily realized + unrealized loss stop	Example research setting: 1.5%-2%; disable new entries after breach.
Maximum correlated exposure	Cap aggregate USD-direction exposure across pairs.
Maximum concurrent positions	2-3 during early deployment.
Stop loss	Mandatory on every live position; volatility/structure based.
No averaging down	Do not add to a losing trade unless a separately tested strategy explicitly permits it.
Martingale/grid	Prohibited.
High-impact event behavior	Dedicated mode or flat; no accidental exposure.
Data outage	Freeze new trading and reconcile existing positions.
Manual emergency stop	One action to cancel pending orders and prevent new ones.

OANDA itself warns that leverage magnifies both profits and losses and that losses can occur rapidly. This blueprint therefore treats leverage as a constraint to manage rather than a target to maximize. [11]

9. Fast Update / Event-Reaction Engine

When a major event occurs, the system should use an event-driven pipeline rather than wait for a slow periodic batch job.

Time	Action
T-30 min	Identify upcoming high-impact release; reduce unrelated strategy aggressiveness; cache consensus and previous values.
T-2 min	Confirm all feeds healthy; record current spread/volatility; prevent stale pending entries.
T0	Receive release/headline timestamp; do not infer from price alone.
T0 + seconds	Parse actual/forecast/previous/revision; classify surprise; update news/event features.

T0 + 1-10 sec	Compare price reaction with expected direction; detect spread shock/slippage risk.
T0 + 10-60 sec	If dedicated event strategy passes all gates, create controlled proposal; otherwise NO TRADE.
Continuous	Re-score thesis whenever significant price/news/macro evidence arrives.

Key rule: Fast does not mean immediate. On major releases, the first seconds can have extreme spread and slippage. The model should optimize expected execution quality, not race blindly to be first.

10. Execution Service

The execution service is deliberately “dumb”: it accepts only a validated TradeIntent from the risk governor. It does not interpret news or override limits.

Function	Requirement
Order creation	Unique client order ID; idempotent retry behavior.
Protection	Attach or immediately create stop-loss; optional take-profit/trailing logic after validation.
Reconciliation	Continuously compare internal positions with broker account state.
Slippage check	Abort/reduce market order if observed execution conditions violate tested limits where order type permits.
Logging	Persist request, response, fill price, spread, latency and broker transaction ID.
Failure behavior	Fail closed: uncertainty or broker/API error cannot create extra exposure.

11. Suggested Technology Stack

Area	Recommended v1
Language	Python 3.12+
API service	FastAPI
Async/streaming	asyncio + httpx/websocket/stream client as appropriate
Dataframes/features	Polars or pandas + NumPy
ML	scikit-learn, XGBoost/LightGBM; PyTorch only where justified
NLP/event extraction	LLM/API or local transformer constrained to a structured schema
Database	PostgreSQL + TimescaleDB extension or PostgreSQL partitioned time-series tables
Cache/message bus	Redis; later Kafka/Redpanda if event volume requires it
Orchestration	Docker Compose initially; cloud containers later
Monitoring	Prometheus/Grafana or cloud equivalent; structured logs

Secrets	Environment/secret manager; never source-control API tokens
Backtesting	Custom event-aware engine or framework validated against broker candle conventions

12. Minimum Data Tables

Table	Purpose
market_prices	Timestamped bid/ask/mid/spread by instrument.
candles	OHLC features by instrument/granularity.
economic_events	Scheduled time, currency/country, event, consensus, prior, actual, revision, importance.
news_events	Publish time, ingest time, source, entities/currencies, event type, sentiment, novelty, confidence.
features	Point-in-time feature vectors keyed by timestamp/horizon.
predictions	Per-component and final calibrated outputs.
trade_intents	Proposed action before risk approval.
risk_decisions	Approved/rejected plus reason and limits.
orders_fills	Broker request/response, fill, slippage, status.
positions_snapshots	Periodic broker-reconciled exposure/equity.
model_registry	Model version, training period, validation statistics and deployment status.

13. Core Real-Time Workflow (Pseudocode)

```

while system_is_enabled:
    event = await event_bus.next()
    update_point_in_time_state(event)

    for pair in affected_currency_pairs(event):
        if not feeds_are_fresh(pair):
            emit(NO_TRADE, reason="stale data")
            continue

        features = build_features(pair, now_utc())
        regime = regime_model.predict(features)
        price_view = price_model.predict(features)
        macro_view = macro_model.predict(features)
        news_view = news_model.predict(features)

        decision = meta_model.combine(
            price_view, macro_view, news_view, regime
        )
        decision = calibrate_and_explain(decision)

        intent = policy_engine.to_trade_intent(decision)
        risk_result = risk_governor.evaluate(intent, account_state)

        if not risk_result.approved:
            emit(NO_TRADE, reason=risk_result.reason)

```

```

elif execution_mode == "paper":
    paper_broker.execute(risk_result.approved_intent)
else:
    oanda_execution.execute(risk_result.approved_intent)

monitor_open_theses_and_positions()

```

14. Testing and Rollout Plan

Phase	Requirement to progress
Phase 0 - Data only	Run feeds continuously; verify timestamps, reconnects, missing data and database integrity. No predictions.
Phase 1 - Historical research	Build point-in-time dataset; establish simple baselines; include realistic transaction costs.
Phase 2 - Walk-forward backtest	Multiple market regimes; no leakage; report drawdown, Sharpe-like risk metrics, hit rate, payoff ratio and calibration.
Phase 3 - Live shadow mode	Model observes live market and records decisions but sends no orders.
Phase 4 - OANDA demo execution	End-to-end streaming, risk, orders, stop logic and reconciliation on demo.
Phase 5 - Small live pilot	Only after predefined stability criteria; smallest practical exposure; strict kill switches.
Phase 6 - Controlled scaling	Increase size based on statistically meaningful live results, never because of one large winning trade.

14.1 Metrics that matter

- Net return after spread/slippage/financing
- Maximum drawdown and drawdown duration
- Expected value per trade
- Profit factor and payoff ratio
- Calibration: do 70% predictions actually win about 70% in comparable conditions?
- Performance by regime and event type
- Tail losses / worst 1% of outcomes
- Average and 95th-percentile decision-to-order latency
- Data-feed uptime and stale-data incidents
- Difference between backtest, shadow, demo and live execution results

15. What the Model Must Never Do

- Promise a fixed daily or monthly return.
- Trade simply because “something is moving.”
- Increase leverage to recover a loss.
- Use martingale or unlimited averaging down.
- Ignore spreads, slippage or stale prices.
- Treat every headline as equally trustworthy.
- Use revised economic data as though the revision was known historically.
- Place an order if the risk engine is unavailable.
- Continue autonomously after position reconciliation fails.

- Optimize solely for win rate; a high win rate can coexist with catastrophic tail losses.

16. Practical Build Roadmap

Sprint	Deliverable
1	OANDA demo account + token; price streamer; candle downloader; account-state reader; PostgreSQL schema.
2	Economic calendar ingestion; official-data adapters; event normalization; UTC timestamp discipline.
3	News ingestion + currency/event classifier; source scoring; novelty detection.
4	Feature engine + baseline trend/volatility/regime models.
5	Macro-surprise and news-event models; opposing-evidence module.
6	Meta-model, probability calibration and BUY/SELL/NO TRADE policy.
7	Independent risk governor, position sizing, portfolio exposure limits and kill switch.
8	OANDA demo execution, order reconciliation and audit logs.
9	Walk-forward evaluation + live shadow comparison; tune only from predefined research protocol.
10	Extended demo period and readiness review before any live-money discussion.

17. Broker Independence

Even though OANDA is the v1 recommendation, define a BrokerAdapter interface so the research and decision system does not depend on OANDA-specific objects. The adapter should expose normalized methods such as `stream_prices()`, `get_candles()`, `account_state()`, `place_order()`, `cancel_order()`, `positions()` and `transactions()`. This allows later comparison with Interactive Brokers or another regulated broker without rewriting the model.

18. Final Design Recommendation

Build this first: OANDA demo + five major FX pairs + OANDA live pricing stream + Trading Economics calendar + official FRED/BLS/ECB validation + structured news sentiment + a three-part ensemble (price, macro, news) + regime detector + independent risk governor. Operate at 15-minute/1-hour/4-hour horizons, not HFT speeds.

The strongest version of this system will not be the one that predicts every price movement. It will be the one that recognizes when it has enough evidence to act, recognizes when conditions have changed, controls exposure when it is wrong, and preserves a complete point-in-time record so every decision can be audited and improved.

19. Sources and Current Technical References

1. OANDA U.S. - Web Trading Platform / v20 REST API overview

<https://www.oanda.com/us-en/trading/platforms/oanda-trade-web/>

2. OANDA U.S. - Account comparison (API trading; variable contract sizes)

<https://www.oanda.com/us-en/trading/account-comparison/>

3. OANDA Developer - Pricing endpoints and pricing stream

<https://developer.oanda.com/rest-live-v20/pricing-ep/>

4. OANDA Developer - Order endpoints

<https://developer.oanda.com/rest-live-v20/order-ep/>

5. Trading Economics - API overview / streaming updates

https://docs.tradingeconomics.com/get_started/

6. Trading Economics - Economic Calendar

https://docs.tradingeconomics.com/economic_calendar/

7. Federal Reserve Bank of St. Louis - FRED API

<https://fred.stlouisfed.org/docs/api/fred/>

8. U.S. Bureau of Labor Statistics - Public Data API

https://www.bls.gov/bls/api_features.htm

9. European Central Bank - Data Portal API

<https://data.ecb.europa.eu/help/api/overview>

10. Alpha Vantage - News & Sentiment API

<https://www.alphavantage.co/documentation/>

11. OANDA U.S. - Forex trading and leverage risk disclosure

<https://www.oanda.com/us-en/trading/forex/>

12. ECB - Data service details and update/delta parameters

<https://data.ecb.europa.eu/help/api/data>

Source status: checked against publicly available documentation in August 2026. Broker/API capabilities, pricing, eligibility, rate limits and terms can change; verify the provider documentation immediately before implementation or live deployment.

Appendix A - Recommended v1 Pair-to-Macro Mapping

Pair	Highest-priority macro drivers
EUR/USD	Fed vs ECB policy; U.S./Euro-area inflation, labor/growth; risk sentiment.
GBP/USD	Fed vs BoE policy; UK CPI/wages/GDP; U.S. macro; political/fiscal news.
USD/JPY	Fed vs BoJ policy; U.S./Japan yields; intervention risk; risk sentiment.
USD/CAD	Fed vs BoC policy; U.S./Canada labor/inflation; oil sensitivity.
AUD/USD	Fed vs RBA policy; China/global growth; commodities; risk sentiment.

Appendix B - Example Trade Proposal

Field	Illustrative paper-mode example
Pair	EUR/USD
Action	BUY
Confidence	0.68 calibrated
Horizon	1 hour
Regime	Post-event directional trend
Evidence	U.S. macro surprise weaker than consensus; USD selling confirmed by price structure; spread normalizing.
Contrary evidence	4-hour trend still partially USD-supportive.
Entry rule	Only after spread returns below configured percentile and price holds confirmation level.
Invalidation	Price closes below event reaction low / macro interpretation contradicted by new information.
Risk	0.25% of paper equity
Decision if conditions fail	NO TRADE